

Computer Graphics – April 11th, 2024

Name: _____

Matriculation Number: _____

Please read the following instructions carefully:

- You have sixty (60) minutes to complete the exam.
- You can answer in German or in English.
- Check that the exam is complete (19 pages on 10 sheets) and readable!
- Please use a blue or black ballpen, not a red or green one. Do not use a pencil!
- Only text on the stapled sheets will be evaluated.
- If needed, we staple additional blank sheets.
- No other material except a calculator is allowed!

I have read and understood the instructions above.

Erlangen, April 11th, 2024

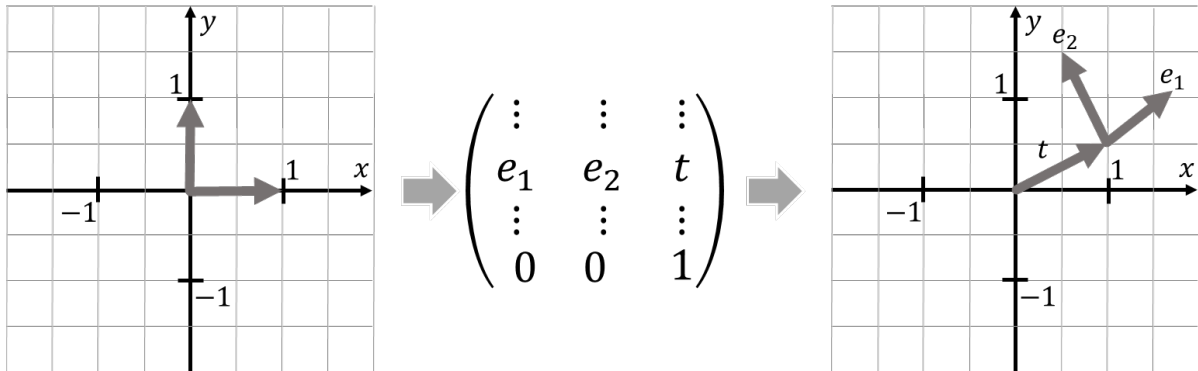
Signature

1	2	3	4	5	6	ALL
12	12	8	12	8	8	60

Grade:

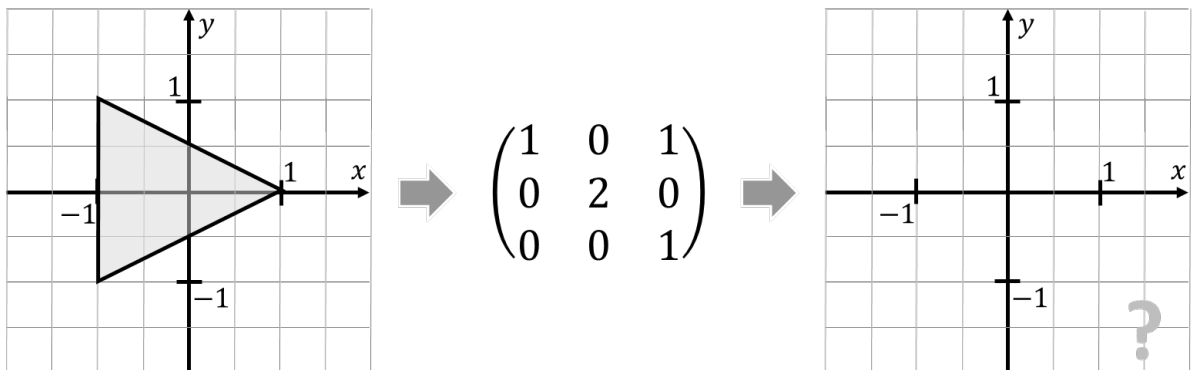
Assignment 1: Affine Transformations [12 points]

Affine mappings in **2D** can be expressed using **homogeneous coordinates** and 3×3 matrices with the following structure, where $(\vec{e}_1, \vec{e}_2)$ is the linear part (image of base vectors) and $\vec{t}$ is the translation:

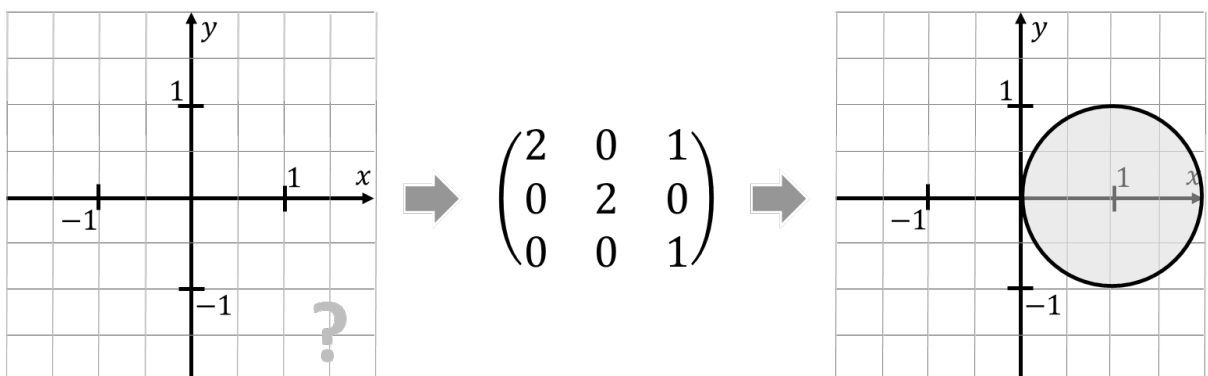


The following illustrations show a shape, which is transformed using an affine matrix. Fill in the missing target image (a), source image (b), and matrix (c).

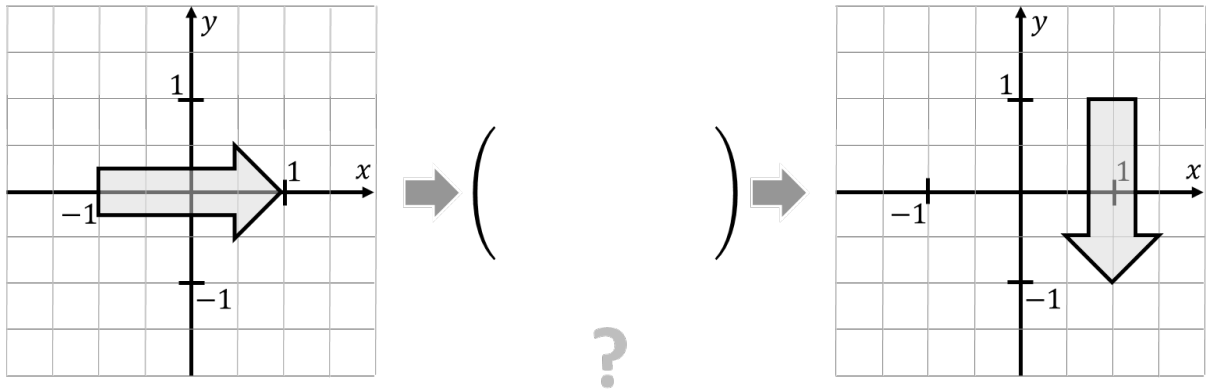
a) [1P]



b) [2P]



c) [2P]



d) [2P] Write down a matrix product to scale an object uniformly by a factor of 2, where the scaling center is at (1,0). Take care to write down the product in the correct order.

$$M = \left(\quad \right) \left(\quad \right) \left(\quad \right)$$

e) [2P] In this task, we will look at **a projective mapping**! Describe the following 2D mapping by a homogeneous 3x3 matrix:

$$(x, y) \rightarrow \left(\frac{x}{x+y}, \frac{y}{x+y} \right)$$

$$M = \left(\quad \right)$$

f) **[3P] Concatenations of transformations.** In the lecture we defined several subclasses of 2D affine transformations. In this assignment, we look at the following classes:

- **affine** transformations
- **rigid** transformations
- **linear** transformations
- **rotations**
- **translations**

We now look at products of matrices of these classes. Given two matrices A and B , each of one of these subclasses. Write down the smallest subclass, to which the product AB belongs.

A	B	$B \cdot A$
rigid	rigid	
translation	rotation	
affine	rigid	
rotation	rotation	
linear	translation	
rigid	translation	

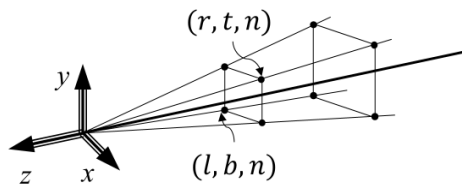
Assignment 2: Perspective [12 points]

a) [2P] Given is a camera with a field of view of 90° and a square image window (aspect ratio 1:1). Write down the perspective matrix assuming a near plane of 1 and a far plane of 10. Below you can find the corresponding lecture slides from the lecture.

$$P = \begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix}$$

- **Perspective Projection:**

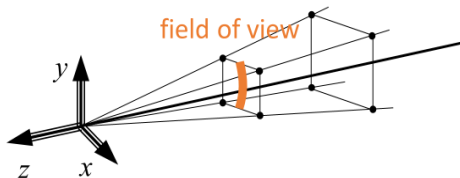
The depth range goes from n to f (near to far), and the image window (l, r, b, t) is defined on the near-plane



- $M_{perspective}(l, r, b, t, n, f) = \begin{pmatrix} \frac{2n}{r-l} & 0 & \frac{l+r}{r-l} & 0 \\ 0 & \frac{2n}{t-b} & \frac{b+t}{t-b} & 0 \\ 0 & 0 & -\frac{f+n}{f-n} & -\frac{2fn}{f-n} \\ 0 & 0 & -1 & 0 \end{pmatrix}$

- Perspective Matrix usually defined by

- opening angle in y-direction: field of view in $y \rightarrow fovy$
- aspect ratio: ration of width over height $\rightarrow aspect$
- near and far plane $\rightarrow n, f$



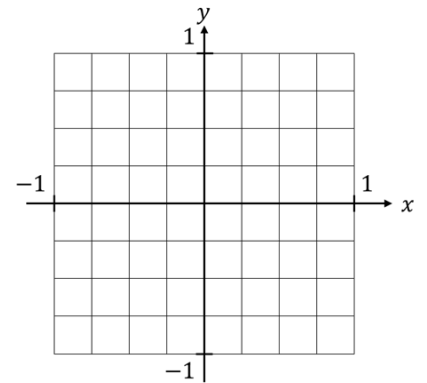
- $-l = r = aspect \cdot n \cdot \tan \frac{fovy}{2}$
- $-b = t = n \cdot \tan \frac{fovy}{2}$

b) **[4P]** Another camera has the following perspective matrix (which is different from a)

$$P = \begin{pmatrix} 0.5 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0 \\ 0 & 0 & -3 & -4 \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

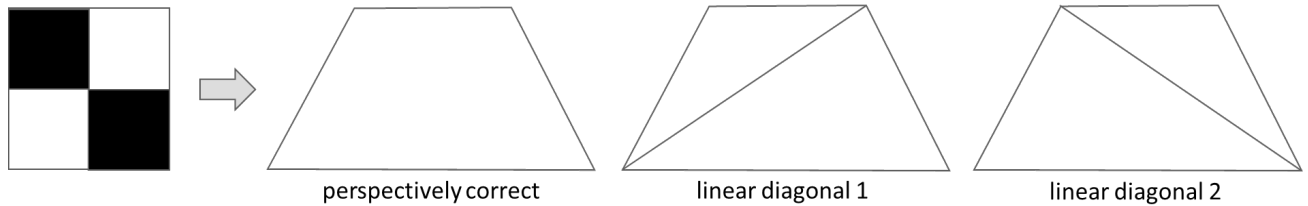
Map the rectangle with vertices $(-1, 1, -1)$, $(-1, 1, -2)$, $(1, 1, -2)$, $(1, 1, -1)$ using above matrix to image coordinates and sketch the 2D result.

$(-1, 1, -1)$	→
$(-1, 1, -2)$	→
$(1, 1, -2)$	→
$(1, 1, -1)$	→



c) **[1P]** The vanishing points of a perspective projection are the points where lines parallel to the main axes converge. They can directly be found as columns in the perspective matrix. How many vanishing points does above matrix P have?

d) **[3P]** Assume, we map a simple 2x2 checker board texture to a square, which is then mapped perspectively as shown below. Show how the texture is mapped to image space using perspectively correct interpolation (left), and with linear interpolation on two triangles, generated by subdivision of the square (right).



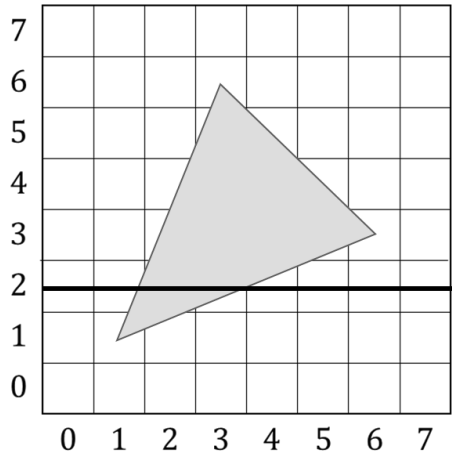
e) **[1P]** We now look at MIP-mapping (without trilinear interpolation). A point on a triangle at camera distance of 10m is rendered with MIP-map-level 3 (level 0 is the level with one single pixel only). Which level will be used, if the camera gets closer and the distance of the point goes down to 5m?

f) **[1P]** In which distance range the coarsest MIP-Map-level 0 will be used?

Assignment 3: Rasterization [8 points]

In this assignment, we want to rasterize triangles using the scanline-algorithm.

a) [3P] Fill in the **edge table** for the following triangle with vertices $A = (1,1)$, $B = (6,3)$, $C = (3,6)$.



y_{lower}	x_{lower}	y_{upper}	$1/m$	$next$

b) [2P] Fill the **active edge table** below for scanline $y_{scan} = 2$

x_{inter}	y_{upper}	$1/m$	$next$

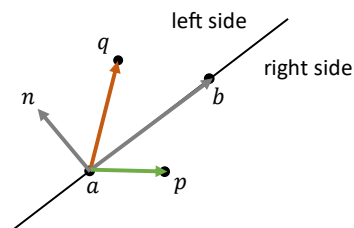
c) [3P] To test whether a point is inside a triangle, we can also check whether the point lies on the “inner” side of each triangle edge. This edge test can be written as a sign test of the scalar product of the point (in homogeneous coordinates) with an edge vector.

Determine this edge vector for **one of the three edges** of the triangle from a). The corresponding lecture slide is added below.

Triangle Test

- Remember: Edge test

- ab defines direction and separates plane to “left” and “right” half
- normal vector n defines these halves:
 $n = \begin{pmatrix} a_2 - b_2 \\ b_1 - a_1 \end{pmatrix}$ points to the left



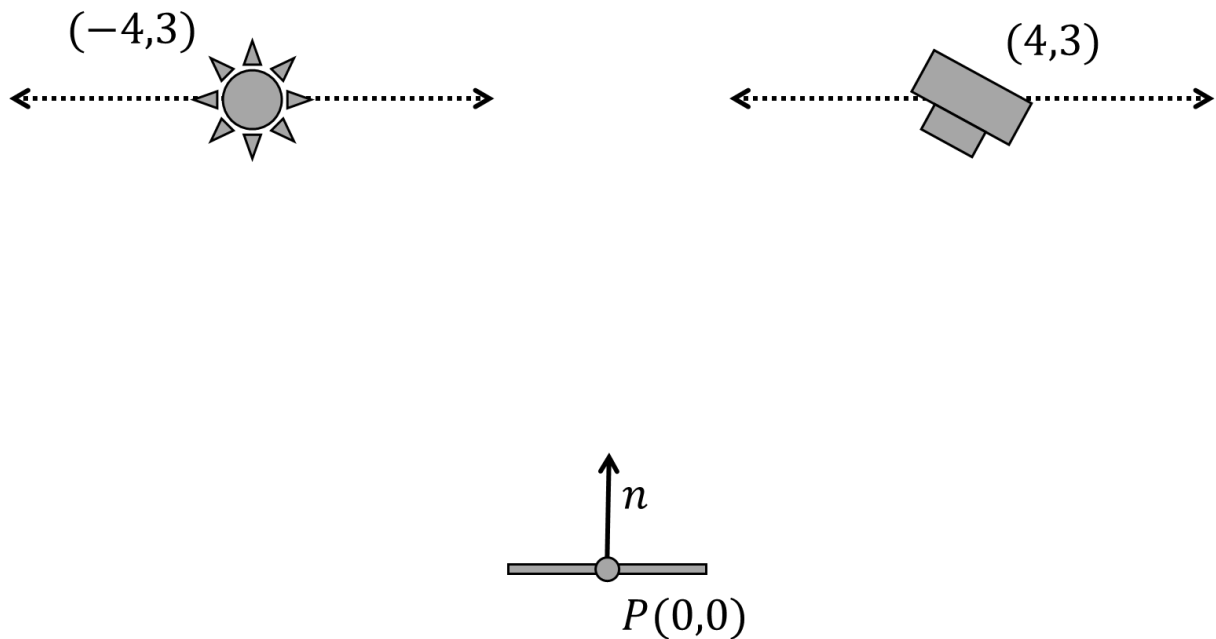
- edge test by using dot product:
 $p \text{ "left"} \Leftrightarrow (p - a) \circ n > 0 \Leftrightarrow p \circ n - a \circ n > 0$
- with homogeneous coordinates:

$$p \text{ "left"} \Leftrightarrow \begin{pmatrix} p_1 \\ p_2 \\ 1 \end{pmatrix} \circ \underbrace{\begin{pmatrix} a_2 - b_2 \\ b_1 - a_1 \\ a_1 b_2 - a_2 b_1 \end{pmatrix}}_{\text{"edge" vector}} > 0$$

→ precompute and use within loop for fast test

Assignment 4: Phong Lighting & Shading [12 points]

The image below shows a point P at $(0,0)$ belonging to a surface with the normal vector $(0,1)$. The surface is lit by a point light located at $(-4,3)$ and captured by a camera located at $(4, 3)$. The light coming from the point light is not attenuated. For subtasks a) and b), you can ignore the dashed arrows.



- a) **[3P]** For the point P , write down the respective light vector l , view vector v and reflection vector r !

- b) **[5P]** Let's assume the following Phong Lighting parameters for the surface:
 $k_{amb} = (0,0,0)$, $k_{diff} = (0,0,1)$, $k_{spec} = (1,0,0)$, $n_{shiny} = 2$
The point light is of intensity $I_{in} = (1,1,1)$.
Compute the color of point P !



- c) **[4P]** Next, assume that both the point light and the camera can move along the dashed lines. When one of the two components moves, the other stays at its starting point. For both the diffuse and the specular part, state what happens for the four movements in the tables below.

First, how does the diffuse part change for each of the following four position changes? In each row, mark one of the possibilities!

		diffuse does not change	diffuse gets darker	diffuse gets brighter
light	moves left			
	moves right			
camera	moves left			
	moves right			

Second, how does the specular part change for each of the following four position changes? In each row, mark one of the possibilities!

		specular does not change	specular gets darker	specular gets brighter
light	moves left			
	moves right			
camera	moves left			
	moves right			

Assignment 5: Run-time Complexity [8 points]

In this assignment we compare run-time complexities of different rendering algorithms.

We assume we want to rasterize a scene with T triangles and V vertices into an image with P pixels. All triangles have roughly the same size of S pixels. We assume that we are in a closed scene, i.e. there are no empty background pixels and every pixel is covered by the scene.

a) [4P] Rasterization:

In the following table, we want to summarize the number of required calls of certain steps of the rendering pipeline. “Lighting Computations” is the number of evaluations of the lighting model (e.g., Phong), assuming a single light source. Note that deferred shading requires two render passes, **in this assignment we only consider the more costly calls from the second lighting pass!**

Fill in the missing fields in the following table:

Algorithm	Vertex Shader Calls	Pixel Shader Calls	Lighting Computations	Depth tests
Rasterization with Gouraud Shading	V			
Rasterization with Pixel Shading		$T \cdot S$		
Rasterization with Deferred Shading (second pass only)	4 (screen-space-aligned quad)			
Rasterization with 4x SSAA			$4 \cdot T \cdot S$	

b) [2P] Ray Casting

If we render the same scene with ray-casting (i.e., with eye rays only). We apply no anti-aliasing, i.e. we have one eye ray per pixel. How many ray-scene intersections and lighting computations are needed then?

#Ray-scene intersections =

#Lighting computations =

c) [2P] Ray Tracing

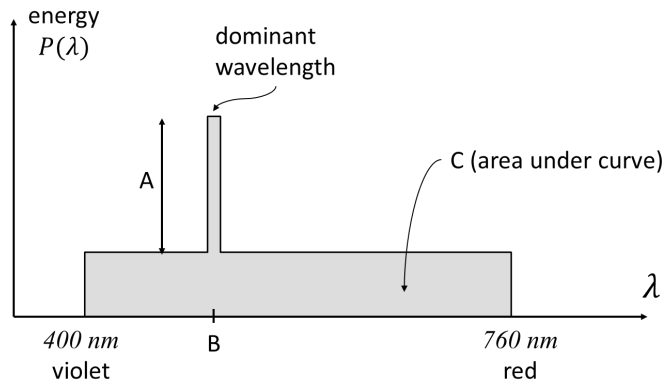
Finally, we render our scene with ray tracing, including reflection and refraction rays, as well as shadow rays to one light source. We apply 4x super-sample anti-aliasing, and a maximum recursion depth of two. How many ray-scene intersections are needed at maximum?

#Ray-scene intersections max. =

Assignment 6: Color Models [8 points]

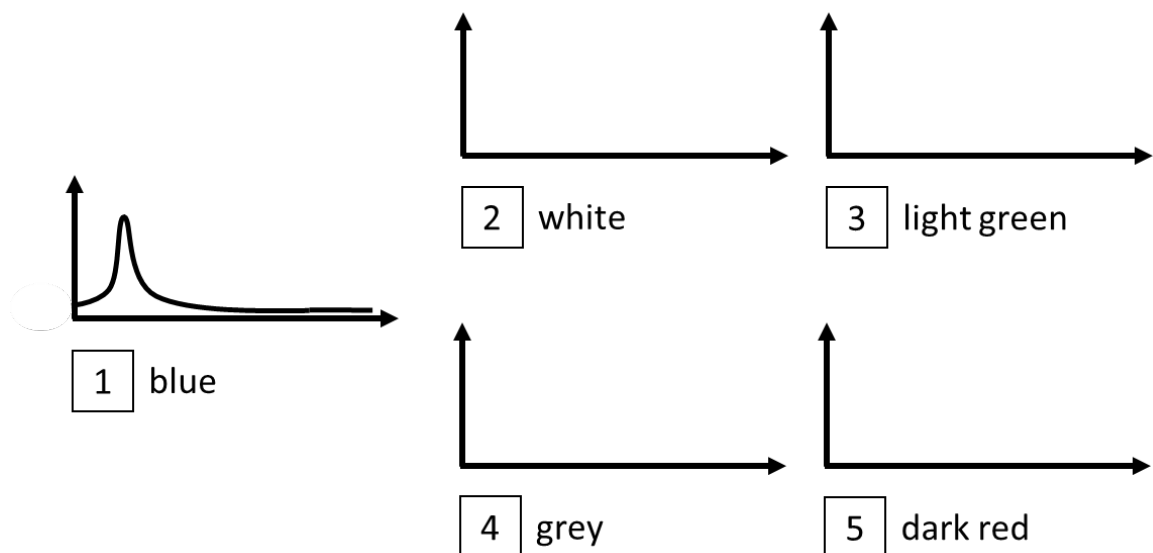
- a) **[2P]** The following figure illustrates the three parameters A , B , C that are often used to define colors: hue, saturation and value. Relate each of the values to A , B , C :

	A, B, or C
Hue	
Saturation	
Value	



- b) **[2P]** Colors can be characterized by their spectrum, a function of wavelengths the visible color is made up of. For the first color below, a representative spectrum is already given. For each of the other colors, sketch a typical spectrum which leads to this color impression!

Hint: The visible spectrum starts from violet (on the left) and features blue, turquoise, green, yellow, orange and red before ending with pink (on the right)!



- c) **[4P]** Below you can find representations of the five colors from a) in each of the three color models RGB, CMYK and HSV. Fill in the respective numbers to link all representations to the correct color. For blue, the solution is already given.

Hint: In HSV, the angles for red, green and blue are 0° , 120° and 240° , respectively.

RGB	CMYK	HSV
1 (0, 0, 1)	(0, 0, 0, 0)	(0° , 0, 0.5)
(0.5, 0, 0)	1 (1, 1, 0, 0)	1 (240° , 1, 1)
(1, 1, 1)	(0, 0, 0, 0.5)	(120° , 0.5, 1)
(0.5, 1, 0.5)	(0, 0.5, 0.5, 0.5)	(0° , 0, 1)
(0.5, 0.5, 0.5)	(0.5, 0, 0.5, 0)	(0° , 1, 0.5)